

Seeking *Principal/Chief System Architect, Enterprise Architect* — I design large-scale distributed systems end-to-end, from reference architecture through production operations, with a track record of delivering mission-critical platforms at state and enterprise scale.

Highlights

“ Enterprise Architecture & Governance

- Authored reference architectures for \$93M state program and \$100M enterprise product, both delivered to production.
- Published architecture papers: *“Lenovo Open Cloud Automation Reference Architecture”*, *“RedHat Hyperconverged Infrastructure (RHII) Reference Architecture”*.
- Expert in architecture decision frameworks, SBOM compliance, FedRAMP (NIST 800-53), and formal state/DIT review processes.

“ Distributed Systems & Cloud-Native Architecture

- Multi-cloud (AWS, Azure): serverless, event-driven, microservices, container orchestration (Kubernetes/OpenShift), API-led connectivity.
- Designed systems spanning 7 AWS accounts, 3 Azure subscriptions, multi-cluster ROSA, 13+ MuleSoft APIs, 12 external integrations.
- HA patterns: auto-scaling, DLQ retry, drift detection, Glacier tiering, failover, network bonding, message queues.

“ AI/ML & Data Architecture

- RAG pipelines (Bedrock Claude, Titan embeddings, OpenSearch Serverless, Guardrails), ETL (Synapse, Athena, Glue), streaming ingestion, data lake/warehouse patterns.

“ Integration & API Architecture

- API-led connectivity (MuleSoft), RESTful, GraphQL, SOAP, SCIM, OAuth2/OIDC/SAML, event-driven (EventBridge, SQS/SNS).
- System integration across government, financial, healthcare, and identity systems.

“ Security Architecture

- Identity: SSO, SCIM provisioning, RBAC, Azure AD/Entra, Keycloak.
- Compliance: FedRAMP SSP, NIST 800-53, SBOM, CVE remediation.
- Infrastructure: KMS, Secrets Manager, WAF, VPC Endpoints, Direct Connect, Transit Gateway.

“ Platform & Infrastructure

- IaC: Terraform, Kustomize, Helm, Ansible, Pulumi.
- GitOps: ArgoCD, GitHub Actions, Semantic Release.
- Observability: Splunk, CloudWatch, Prometheus, Grafana.

Experience

12/2023

-

Department of Health and Human Service, North Carolina, Principal System Architect

- “ As the principal system architect of the PATH NC (\$93M) child welfare modernization, led the successful go-live (June 2025) of a statewide platform serving 3,000+ users across all 100 NC counties. Oversaw 50+ repositories, 30+ SaaS/COTS products, 7 AWS accounts, 3 Azure subscriptions, and a multi-cluster OpenShift platform.
- “ Established architecture governance from scratch: authored the reference architecture, formal DIT/State architecture review decision points, custom code dev/build/deploy standards, SBOM and license compliance, backup & restore strategy, production incident management, and PROD health dashboards. Systematically drove go-live readiness and post-go-live tech debt resolution.
- “ Designed the complete system topology: Salesforce application tier (2,745+ Apex classes, 1,195+ LWC, 647+ objects), MuleSoft integration layer (13+ APIs to 12 external systems), serverless document management (Lambda, EventBridge, DynamoDB, S3), GenAI/RAG platform (Bedrock, OpenSearch), data warehouse (Azure Synapse), and container platform (ROSA/OpenShift with ArgoCD).
- “ Architected the user identity and access model: Azure AD/NCID SSO with SCIM provisioning, JIT activation, per-county role matrix (400+ application roles), Salesforce ACL model, and phased rollout validation tooling.
- “ Designed the end-to-end financial data movement architecture (Salesforce → Azure Pipeline → Synapse → Power BI → AWS SFTP → NCFS mainframe) with Power Automate orchestration for monthly county payment processing.
- “ Architected the serverless document management system replacing IBM FileNet: event-driven ingestion, intelligent document processing (Textextract, Bedrock), DLQ retry orchestration, Glacier archival, KMS encryption, JWT authorization, VPC Endpoints.
- “ Designed the GenAI platform architecture (GenAIE): evaluated alternatives (Kendra, Salesforce AgentForce, Microsoft Copilot), selected RAG approach with Bedrock Claude/Titan, OpenSearch Serverless, Bedrock Guardrails, deployed on OpenShift across 6 environments.
- “ Defined infrastructure architecture: multi-cluster ROSA with auto-scaling (up to 12 nodes/96 CPU/768 GB), Terraform IaC, Kustomize/ArgoCD GitOps, drift detection, Day 0/1/2 automation, network design across Direct Connect, Transit Gateway, Cloudflare.
- “ Led security architecture: FedRAMP SSP (NIST 800-53), secrets lifecycle management, network segmentation, security scanning pipeline (Trivy, Veracode, SonarQube), and compliance tracking.

3/2022

-

Lucidum, Principal System Architect

- “ Led design of the next-gen product architecture transforming on-prem-only model to hybrid and cloud-native. Introduced AWS & Azure services achieving successful transition of twelve production systems.
- “ Defined the system's multi-tenant architecture spanning Python, Java, and React services deployed via Helm/K8S with Jenkins and ArgoCD. Oversaw

50+ repositories across the full stack.

- “ Architected CI/CD pipeline system: multi-branch building, matrix testing, A/B deployment, GitHub Actions/Package integration, CVE scanning, and secret management (Hashicorp Vault, AWS/Azure).
- “ As AWS and Azure administrator: SSO, IAM, ECR, ECS, RDS, Aurora, Lambda, CloudWatch, VPC, WAF, ELB, Route53, CloudFront.

11/2016 Lenovo US, P8, Advisory Engineer, Senior Solution Architect, Team Lead
- 3/2022

- “ As the principal system architect and first author of the “*Lenovo Open Cloud Automation Reference Architecture*”. Designed a baremetal-to-cloud orchestration platform from concept to market-ready product in 13 months. Showcased at MWC Barcelona 2019, won \$100M contract.
- “ As the principal system architect and first author of the “*RedHat Hyper-converged Infrastructure (RHHI) Reference Architecture*”. Designed a 3-12 server VM-workload solution with Glusterfs, layer-3 networking, HA, fault tolerance, and zero-touch provisioning.
- “ As the principal architect of the “*Lenovo Workload Solution Store*”, a baremetal orchestration solution using declarative taxonomy, serving as the core foundation of the ThinkAgile VX product.
- “ As team lead of ThinkAgile CP, a hybrid cloud platform with on-prem infrastructure and AWS management plane. Led global team delivering six production releases over two years.
- “ Designed a DCIM system (Netbox-based) administrating 300+ servers and switches globally: Django, Celery, Redis, Docker, Ansible.

01/2015 PY Consulting, Founder, Freelance
- 11/2016

- “ Designed and delivered systems across healthcare (UNC, MGH), international development (World Bank), e-commerce (Wei Fashion), and EdTech (Shang Xue Tang). Consistently bootstrapped architecture, SDLC, and DevOps practices from zero for each engagement.

04/2004 Bit 9 Inc. (Carbon Black), Researcher, Tech Lead
- 05/2007

- “ Led research securing the Windows NT kernel, resulting in \$6M A-round from Kleiner Perkins.

07/1998 Instron Corp., Senior Engineer
- 04/2004

- “ Architected real-time firmware systems: VxWorks RTOS, real-time Linux, C, DSP, PLC, DAQ.

Education

- “ *MBA (04/2007 - 09/2008)*, Boston Univ., International Management & Finance
- “ *Master of Science (09/1995 - 05/1997)*, Ohio Univ., Electrical Engineering
- “ *Bachelor of Science (09/1990 - 07/1994)*, Shanghai JiaoTong Univ., China, Electrical Engineering

Publications

1. <https://lenovopress.com/lp1148-red-hat-hyperconverged-infrastructure-for-virtualization-reference-architecture>
2. <https://lenovopress.com/lp1149-lenovo-open-cloud-reference-architecture>